7CS033 – Data Mining and Informatics

Instructions to Install Software

This guide is applicable for Windows computer for installing Python 3.x, checking PIP3 exist on your computer, installing Jupyter Notebook, and installing external Python packages.

If you are using your personal computer, this guide will be helpful in providing information about the tools required for this module and how to instal them.

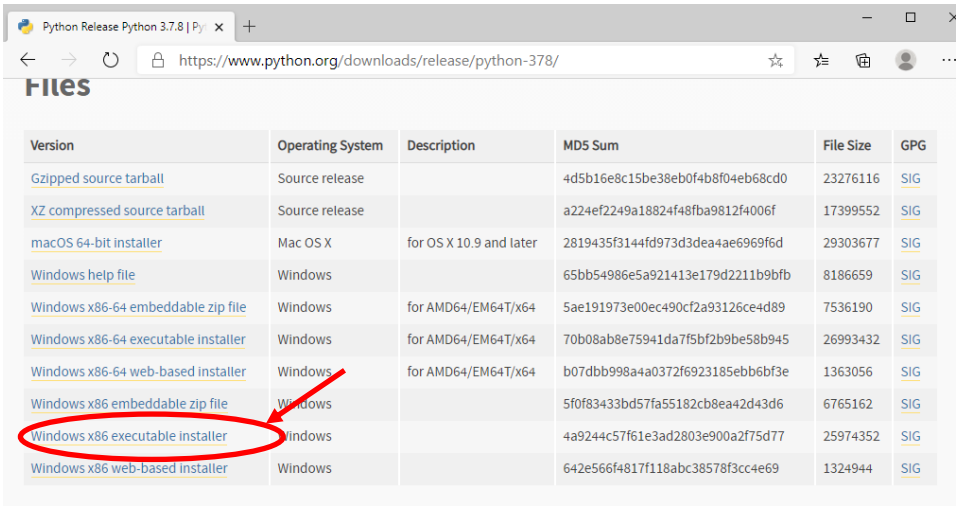
If you are using University's system (e.g., working in the lab), these tools will be accessible via Apps Anywhere. How to use them, will be demonstrated during the teaching session (Workshops).

The following tools/installations are required for the module:

1. Installing Python 3.7
2. Checking pip3
3. Installing Jupyter Notebook
4. Installing any Python-based external libraries or packages
5. Installing Notepad++

1. Installing Python 3.7

a. Open the link https://www.python.org/downloads/release/python-378/ scroll down to locate "**x86–64 executable installer**" and click to download it.



Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		4d5b16e8c15be38eb0f4b8f04eb68cd0	23276116	SIG
XZ compressed source tarball	Source release		a224ef2249a18824f48fba9812f4006f	17399552	SIG
macOS 64-bit installer	Mac OS X	for OS X 10.9 and later	2819435f3144fd973d3dea4ae6969f6d	29303677	SIG
Windows help file	Windows		65bb54986e5a921413e179d2211b9bfb	8186659	SIG
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64T/x64	5ae191973e00ec490cf2a93126ce4d89	7536190	SIG
Windows x86-64 executable installer	Windows	for AMD64/EM64T/x64	70b08ab8e75941da7f5bf2b9be58b945	26993432	SIG
Windows x86-64 web-based installer	Windows	for AMD64/EM64T/x64	b07dbb998a4a0372f6923185ebb6bf3e	1363056	SIG
Windows x86 embeddable zip file	Windows		5f0f83433bd57fa55182cb8ea42d43d6	6765162	SIG
Windows x86 executable installer	Windows		4a9244c57f61e3ad2803e900a2f75d77	25974352	SIG
Windows x86 web-based installer	Windows		642e566f4817f118abc38578f3cc4e69	1324944	SIG

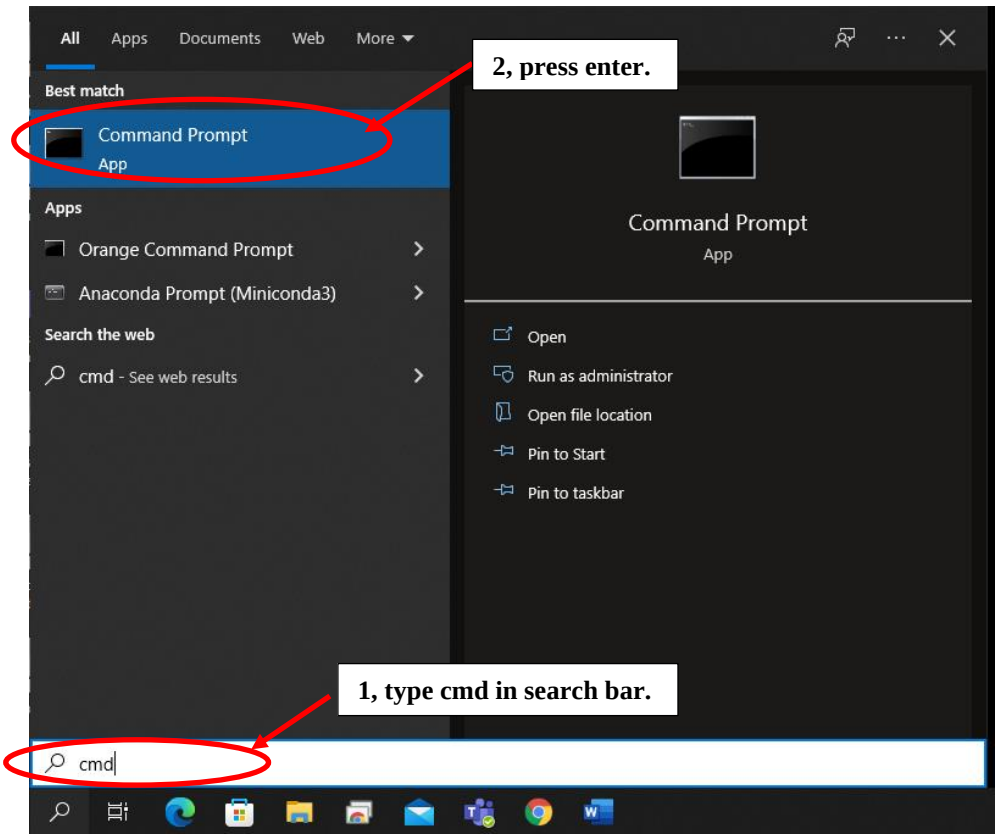
b. Locate "**x86–64 executable installer",** it usually goes in the Download folder, and double-click to install the program.

c. On the installation window, tick the last option too. i.e., '**Add python 3.7 to PATH**'. And, install Python by selecting Install Now.

d. Once it is installed successfully, use the command shown in the image to check the version. This command will run in command prompt. If you are not sure how to open cmd, please see next point ‘e.’.

```
C:\Users\Vinita>python --version
Python 3.7.2
```

e. To open command prompt, on the start menu, type cmd and press enter. Command prompt will be opened. Then type command as shown in the image below.



2. Checking pip3

a. PIP is a Python package manager or installer for python. The Python 3 installers for Windows will install pip3 automatically. Just to confirm that it exists, type 'where pip3' in the command prompt as shown in the image below.

```
C:\Users\Vinita>where pip3
C:\Users\Vinita\AppData\Local\Programs\Python\Python37\Scripts\pip3.exe
```

b. pip3 will be required to install python-based external machine learning libraries or packages. Command to install external libraries are provided in the later section of this document.

3. Installing Jupyter Notebook

a. To install jupyter notebook, type '**pip3 install -U notebook**' in the command prompt as shown in the image below.

```
C:\Users\Vinita>pip3 install -U notebook
```

For additional information, please check the link - https://jupyter.org/install

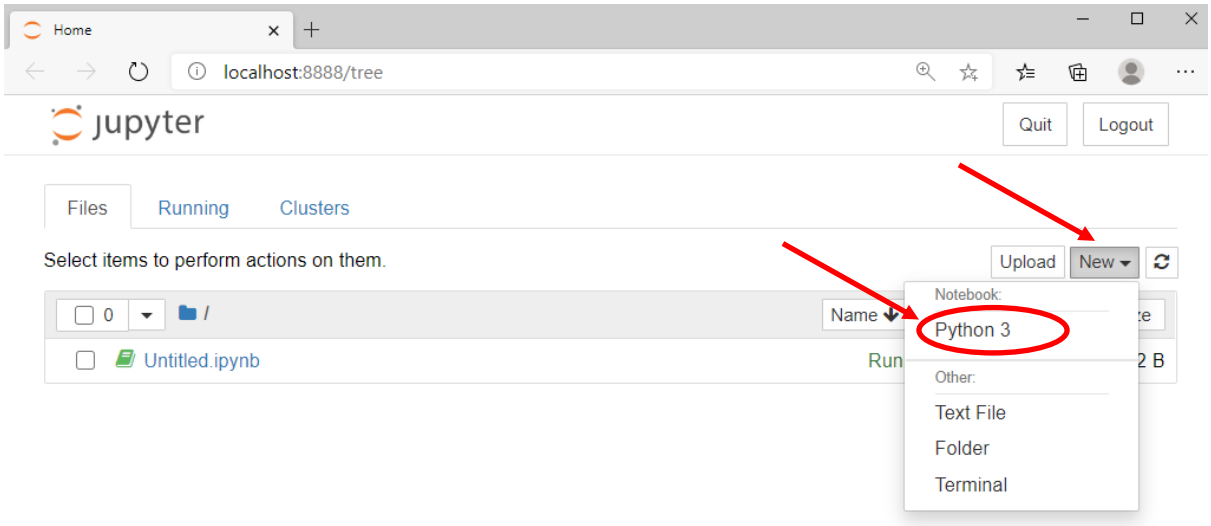
However, please note that you will need to use command **pip3 install -U notebook**.

4. We are now ready to run Jupyter Notebook

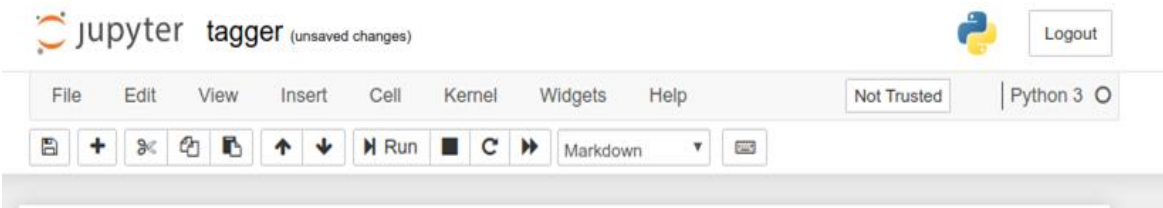
Open command prompt, then on command prompt type **jupyter notebook** as shown in the image below. **NOTE:** it is jupyter NOT jupiter.

```
C:\Users\Vinita>jupyter notebook
```

This will open the Jupyter web interface, where you can create your first jupyter notebook by clicking on 'New' and selecting 'Python 3'. Or, you can select the existing notebook if you have one already.



5. Working with a notebook



Once you have opened a notebook, you will see the toolbar at the top of the screen.

Using the toolbar, you will be able to carry out the most essential tasks. For instance, you may wish **to run the notebook**. This can be done by clicking on *Kernel* and then on *Restart & Clear Output.* You will then go through each cell and run it individually. In order to do this, select the cell you wish to run by clicking on it and pressing *Run* at the top of the screen. Remember, many notebooks rely on the cells being run in sequence (from top to bottom), so running single cells without the previous ones could result in errors.

Editing the notebook is as simple as selecting a cell and editing it just as you would a normal text file. In the workshop tasks, you may be given questions to answer in the notebooks. To do so, select the empty cell below the cell which contains the question. If there is not an empty cell, you can insert one by selecting a cell beneath which you would like to insert a new cell. Then simply click on the + button on the toolbar.

Remember, cells can be of **different types**. You can find out what type your cell is by selecting it, and the checking the text box on the toolbar which is close to the *Run* button. You should see something like *markdown* or code. In order to write plain text, make sure your cell type is *markdown*. If you wish to add code, make sure you have selected *code.*

6. Installing packages/modules

Packages or **modules** are parts of software that add functionality to Python. Before using modules and packages you may have to add them. This could be because you have been instructed to do so, you have come across an error indicating that one is missing, or you have chosen to solve a task by using functionality from a different package.

To install a package, you will have to start the terminal/command prompt. **Please make sure you use a new window and not the one where you started the jupyter notebook. It is also recommended to close Jupyter and restart it once the package has been installed.** Once you have an open Terminal window, you can use the Python package manager called *pip3*. Installing a package is simple. Command to instal a package is:

pip3 install -U *package name*

Let's assume you would like to install the *pandas* package: in command prompt, type in '**pip3 install -U pandas'.**

```
C:\Users\Vinita>pip3 install -U pandas
```

7. Saving a notebook as a PDF

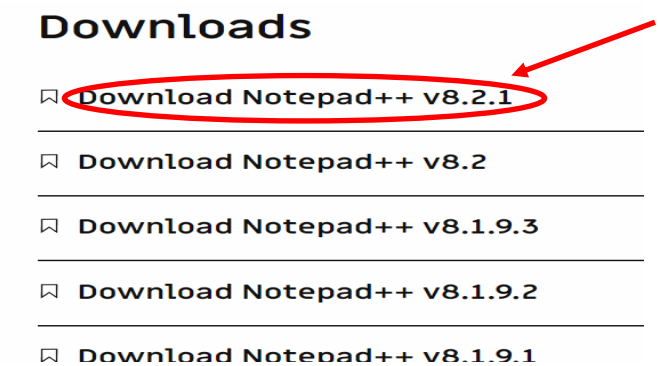
To hand in your work, you will be asked to upload a PDF of your notebook via Canvas. Creating a PDF of your notebook is simple: With your notebook open in Jupyter, press **Ctrl** + **P**, and you should be shown a prompt to print the page. From the available printers (if any), you should be able to select *PDF* or *Send to file*. Once you have selected this, press *print* or *ok* and you should be prompted to save the file somewhere.

Please note that you are always expected to go through the PDF you have produced and double check whether it is readable. This includes sections where you are displaying data or output by running a command or answers, discussion, comments etc. If something is not readable or displayed incorrectly, please go back to the notebook and make the appropriate changes. The way the notebook is displayed in Jupyter will probably be quite close to how it is exported as PDF, so make sure everything is clearly readable.

8. Installing Notepad++

a. Go to the link - https://notepad-plus-plus.org/downloads/

b. Download on of the versions.



c. Locate (usually, Downloads folder) and Run the installed (.exe) file.

d. You may need Notepadd++ to inspect files of different formats including .csv, python script files.